Meng Gao

Email: mmgao2@hkbu.edu.hk

Date and Place of Birth:
18/09/1991, Nanjing, China

Professional Experience

2019 to present Assistant Professor, Hong Kong Baptist University
2016 to 2019 Postdoctoral Fellow, Harvard University

Education

2015 Ph.D. in Chemical Engineering, University of Iowa, Iowa City, IA, USA
2015 M.Eng. in Chemical Engineering, University of Iowa, Iowa City, IA, USA
2012 B.S. in Atmospheric Physics, Nanjing University of Information Science & Technology, Nanjing, China

Awards

2020 Atmospheric Sciences Division Outstanding Early Career Scientist Award, European Geosciences Union
2018 The Global Atmosphere Watch (GAW) Early Career Travel Grants, World Meteorological Organization
2017 Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XIV), Department of Energy, USA (25 globally and biennially)
2016 International Global Atmospheric Chemistry Early Career Travel Grants
2016 Karl Kammermeyer Outstanding Research Award, University of Iowa, USA
2015 Graduate College Post-Comprehensive Research Award, University of Iowa, USA
2012 Karl Kammermeyer Fellowship, University of Iowa, USA

Membership on Committees

2020-2022 World Meteorological Organization, Global Air Quality Forecasting and Information System (GAFIS) Steering Committee

Publications

2020

[57] Yang, B.Y., Hu, L.W., Jalaludin, B., Knibbs, L.D., Markevych, I., Heinrich, J., Bloom, M.S., Morawska, L., Lin, S., Jalava, P., Roponen, M., **Gao, M.**, Chen, D.H., Zhou, Y., Yu, H., Liu, R., Zeng, X., Zeeshan, M., Guo, Y., Yu, Y. and Dong, G. Association Between Residential Greenness, Cardiometabolic Disorders, and Cardiovascular Disease Among Adults in China. JAMA Network Open, 3(9), pp.e2017507-e2017507, 2020.

[56] Yu, H., Guo, Y., Zeng, X., **Gao, M.**, Yang, B.Y., Hu, L.W., Yu, Y., Dong, G.H., Zhou, Y., Qian, Z. and Sun, J., 2020. Modification of caesarean section on the associations between air pollution and childhood asthma in seven Chinese cities. Environmental Pollution, 267, p.115443, 2020.

[55] Jia, B., Wang, Y.*, Wang, C., Zhang, Q., **Gao, M.*** and Yung, K.L.L: Sensitivity of $PM_{2.5}$ to NOx emissions and meteorology in North China based on observations. Science of the Total Environment, in press, 2020.

[54] Wang, Y., Wild, O., Chen, X., Wu, Q., Gao, M., Chen, H., Qi, Y. and Wang, Z. Health impacts of long-term ozone exposure in China over 2013–2017. Environment International, 144, p.106030, 2020.

[53] Liu, L., Zhang, X., Xu, W., Liu, X., Li, Y., Wei, J., Gao, M., Bi, J., Lu, X., Wang, Z. and Wu, X. Challenges for global sustainable nitrogen management in agricultural systems. Journal of Agricultural and Food Chemistry, 68(11), pp.3354-3361., 2020.

[52] Saide, P. E., **Gao, M.**, Lu, Z., Goldberg, D., Streets, D. G., Woo, J.-H., Beyersdorf, A., Corr, C. A., Thornhill, K. L., Anderson, B., Hair, J. W., Nehrir, A. R., Diskin, G. S., Jimenez, J. L., Nault, B. A.,

Campuzano-Jost, P., Dibb, J., Heim, E., Lamb, K. D., Schwarz, J. P., Perring, A. E., Kim, J., Choi, M., Holben, B., Pfister, G., Hodzic, A., Carmichael, G. R., Emmons, L., and Crawford, J. H.: Understanding and improving model representation of aerosol optical properties for a Chinese haze event measured during KORUS-AQ, Atmospheric Chemistry and Physics, https://doi.org/10.5194/acp-2019-1022, in press, 2020.

[51] Wang, H., Li, J., **Gao, M.**, Chan, T.C., Gao, Z., Zhang, M., Li, Y., Gu, Y., Chen, A., Yang, Y., and Ho, H.C.: Spatiotemporal variability in long-term population exposure to PM2.5 and lung cancer mortality attributable to PM2.5 across the Yangtze River Delta (YRD) region over 2010 - 2016: A multistage approach. Chemosphere, 10.1016/j.chemosphere.2020.127153, 2020.

[50] Liu, K., Yang, B.Y., Guo, Y., Bloom, M.S., Dharmage, S.C., Knibbs, L.D., Heinrich, J., Leskinen, A., Lin, S., Morawska, L., Jalaludin, B., Markevych, I., Jalava, P., Komppula, M., Yu, Y., **Gao, M.**, Zhou, Y., Yu, H.Y., Hu, L.W., Zeng, X.W. and Dong, G.H.: The role of influenza vaccination in mitigating the adverse impact of ambient air pollution on lung function in children: new insights from the Seven Northeastern Cities Study in China. Environmental Research, p.109624, 2020.

[49] Lu, X., Zhang, L., Wang, X., **Gao, M.**, Li, K., Zhang, Y., Yue, X. and Zhang, Y.: Rapid increases in warm-season surface ozone and resulting health impact over China since 2013. Environmental Science & Technology Letters, 7, 4, 240-247, 2020.

[48] Shen, L., Hao, F., **Gao, M.**, Wang, H., Zhu, B., Gao, J., Cheng, Y. and Xie, F.: Real-time geochemistry of urban aerosol during a heavy dust episode by single-particle aerosol mass spectrometer: Spatio-temporal variability, mixing state and spectral distribution. Particuology, https://doi.org/10.1016/j.partic.2020.02.001, 2020.

[47] Wang, Y., Wild, O., Chen, H., **Gao, M.**, Wu, Q., Chen, X., and Wang, Z.: Acute and chronic health impacts of PM2.5 in China and the influence of interannual meteorological variability, Atmospheric Environment, 229, 117397, 2020.

[46] **Gao, M.,** Gao, J., Zhu, B., Kumar, R., Lu, X., Song, S., Zhang, Y., Jia, B., Wang, P., Beig, G., Hu, J., Ying, Q., Zhang, H., Sherman, P., and McElroy, M. B.: Ozone Pollution over China and India: Seasonality and Sources, Atmospheric Chemistry and Physics, 20, 4399–4414, https://doi.org/10.5194/acp-20-4399-2020, 2020.

[45] Sun, S., Jin, J., Xia, M., Liu, Y., **Gao, M.**, Zou, C., Wang, T., Liu, Y., Wu, L., Mao, H., and Wang, P.: Vehicle emissions in a middle-sized city of China: current status and future trends, Environmental International, 137, 105514, 2020.

[44] Tao, Z., Chin, M., **Gao, M.**, Kucsera, T., Kim, D., Bian, H., Kurokawa, J., Wang, Y., Carmichael, G. R., Wang, Z., and Akimoto, H.: Evaluation of NU-WRF Performance on Air Quality Simulation under Various Model Resolutions – An Investigation within Framework of MICS-Asia Phase III, Atmos. Chem. Phys., 20, 2319–2339, https://doi.org/10.5194/acp-20-2319-2020, 2020.

[43] Zeng, Z., Wang, Z., Gui, K., Yan, X., **Gao, M.**, Luo, M., Geng, H., Liao, T., Li, X., An, J., Liu, H., He, C., Ning, G., **Yang, Y.**: Daily Global SolarRadiation in China Estimated from High-density Meteorological Observations: A Random Forest Model Framework, ***Earth and Space Science***, in press, 2020.

[42] Fu, P., **Gao, M.** and Yung, K. KL: Association of intestinal disorders with Parkinson's disease and Alzheimer's disease: A systematic review and meta-analysis, *ACS Chemical Neuroscience*, 11, 3, 395-405, 2020.

[41] **Gao, M.**, Liu, Z., Zheng, B., Ji, D., Sherman, P., Song, S., Xin, J., Liu, C., Wang, Y., Zhang, Q., Wang, Z., Carmichael, G., and McElroy, M.: China's emission control measures have suppressed unfavorable influences of climate on wintertime $PM_{2.5}$ concentrations in Beijing since 2002, *Atmospheric Chemistry and Physics*, 20, 1497–1505, https://doi.org/10.5194/acp-20-1497-2020, 2020.

[40] **Gao, M.**, Han, Z., Tao, Z., Li, J., Kang, J.-E., Huang, K., Dong, X., Zhuang, B., Li, S., Ge, B., Wu, Q., Lee, H.-J., Kim, C.-H., Fu, J. S., Wang, T., Chin, M., Li, M., Woo, J.-H., Zhang, Q., Cheng, Y., Wang, Z., and Carmichael, G. R.: Air Quality and Climate Change, Topic 3 of the Model Inter-Comparison Study for Asia Phase III (MICS-Asia III), Part II: aerosol radiative effects and aerosol feedbacks. *Atmospheric Chemistry and Physics*, 20, 1147–1161, https://doi.org/10.5194/acp-20-1147-2020, 2020.

[39] Xu, Y., Wu, X., Kumar, R., Barth, M., Diao, C., **Gao, M.**, Lin, L., Jones, B., and Meehl, G. A.: Substantial increase in the joint occurrence and human exposure of Heatwave and High-PM Hazards (HHH) over South Asia in the mid-21st century, *AGU Advances*, 1, e2019AV000103.https://doi.org/10.1029/2019AV000103, 2020.

[38] Yang, B.Y., Liu, K.K., Markevych, I., Knibbs, L.D., Bloom, M.S., Dharmage, S.C., Lin, S., Morawska, L., Heinrich, J., Jalaludin, B., **Gao, M.**, Guo, Y., Zhou, Y., Huang, W., Yu, H., Zeng, X., Hu, L., Hu, Q., Dong, G.: Association between residential greenness and metabolic syndrome in Chinese adults. *Environment International*, 135, p.105388, 2020.

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2019

[36] **Gao, M.**, Sherman, P., Song, S., Yu, Y., Wu, Z. and McElroy, M. B.: Seasonal Prediction of Indian Wintertime Aerosol Pollution using the Ocean Memory Effect. ***Science Advances***, 5(7), eaav4157, 2019.

[35] Sherman, P., **Gao, M.**, Song, S., Ohiomoba, P., Archibald, A., McElroy, M.: The Influence of Dynamics and Emissions Changes on China's Wintertime Haze, *Journal of Applied Meteorology and Climatology*, https://doi.org/10.1175/JAMC-D-19-0035.1, 2019.

[34] Ji, D., **Gao, M.**, Maenhaut, W., He, J., Wu, C., Cheng, L., Gao, W., Sun, Y., Sun, J., Xin, J., Wang, L., Wang, Y.: The Carbonaceous Aerosol Levels still Remain A Challenge in the Beijing-Tianjin-Hebei region of China: insights from continuous high temporal resolution measurements in multiple cities. *Environmental International*, 126, 171-183, 2019.

[33] Song, S., **Gao, M.**, Xu, W., Sun, Y., Worsnop, D., Jayne, J., Zhang, Y., Zhu, L., Li, M., Zhou, Z., Cheng, C., Lv, Y., Wang, Y., Peng, W., Xu, X., Lin, N., Wang, Y., Wang, S., Munger, J. W., Jacob, D., McElroy, M.B.: Possible Heterogeneous Hydroxymethanesulfonate (HMS) Chemistry in Northern China Winter Haze and Implications for Rapid Sulfate Formation. *Atmospheric Chemistry and Physics*, 19, 1357-1371, 2019.

[32] Yue, S., Bikkina, S., **Gao, M.**, Barrie, L.A., Kawamura, K. and Fu, P.: Sources and Radiative Absorption of Water-Soluble Brown Carbon in the High Arctic Atmosphere. *Geophysical Research Letters*, https://doi.org/10.1029/2019GL085318, 2019.

[31] Song, S., Nenes, A., **Gao, M.**, Zhang, Y., Liu, P., Shao, J., Ye, D., Xu, W., Lei, L., Sun, Y. and Liu, B.: Thermodynamic Modeling Suggests Declines in Water Uptake and Acidity of Inorganic Aerosols in Beijing Winter Haze Events during 2014/2015–2018/2019. *Environmental Science & Technology Letters*, 6(12), pp.752-760, 2019.

[30] Zhou, W., **Gao, M.,** He, Y., Wang, Q., Xie, C., Xu, W., Zhao, J., Du, W., Qiu,Y., Lei, L., Fu, P., Wang, Z., Worsnop, D.R., Zhang, Q., Sun, Y.: Response of Aerosol Chemistry to Clean Air Action in Beijing, China: Insights from two-year ACSM measurements and model simulations, *Environmental Pollution*, https://doi.org/10.1016/j.envpol.2019.113345, 2019.

[29] Cui, Y., Ji, D., Chen, H., **Gao, M**., Maenhaut, W., He, J. and Wang, Y.: Characteristics and Sources of Hourly Trace Elements in Airborne Fine Particles in Urban Beijing, China. *Journal of Geophysical Research: Atmospheres*, https://doi.org/10.1029/2019JD030881, 2019.

[28] Liu, Z., Hu, B., Ji, D., Cheng, **M., Gao**, W., Shi, S., Xie, Y., Yang, S., Gao, M., Fu, H. and Chen, J.: Characteristics of Fine Particle Explosive Growth Events in Beijing, China: seasonal variation, chemical evolution pattern and formation mechanism. *Science of The Total Environment*, 687, 1073-1086, 2019.

[27] Goldberg, D., Saide, P., Lamsal, L., de Foy, B., Lu, Z., Woo, J., Kim, Y., Kim, J., **Gao, M.**, Carmichael, G. R. and Streets, D.: A Top-down Assessment using OMI NO_2 Suggests an Underestimate in the NOx Emissions Inventory in Seoul, South Korea during KORUS-AQ. *Atmospheric Chemistry and Physics*, 19(3), 1801-1818, 2019.

[26] Zhou, M., Zhang, L., Chen, D., Gu, Y., Fu, T., **Gao, M.**, Zhao, Y., Lu, X. and Zhao, B.: The Impact of Aerosol-Radiation Interactions on the Effectiveness of Emission Control Measures. *Environmental Research Letters*, 14, 024002, 2019.

[25] Lu, X., Zhang, L., Zhao, Y., Jacob, D.J., Hu, Y., Hu, L., **Gao, M.**, Liu, X., Petropavlovskikh, I., McClure-Begley, A. and Querel, R.: Surface and Tropospheric Ozone Trends in the Southern Hemisphere since 1990: Possible Linkages to Poleward Expansion of the Hadley Circulation. *Science Bulletin*, 64(6), 400-409, 2019.

2018

[24] **Gao, M.**, Ding, Y., Song, S., Lu, X., Chen, X. and McElroy, M. B.: Secular Changes in Wind Energy Potential in India Associated with Warming Indian Ocean. ***Science Advances***, 4(12), p.eaat5256, 2018.

[23] **Gao, M.**, Beig, G., Song, S., Zhang, H., Hu, J., Ying, Q., Liang, F., Liu, Y., Wang, H., Lu, X., Zhu, T., Carmichael, G. R., Nielsen, C. P. and McElroy, M. B.: The Impact of Power Generation Emissions on Ambient $PM_{2.5}$ Pollution and Human Health in China and India. *Environmental International*, 121, 250-259, 2018.

[22] **Gao, M.**, Ji, D., Liang, F. and Liu, Y.: Attribution of Aerosol Direct Radiative Forcing in China and India to Emitting Sectors. *Atmospheric Environment*, 190, 35-42, 2018.

[21] **Gao, M.**, Han, Z., Liu, Z., Li, M., Xin, J., Tao, Z., Li, J., Kang, J.E., Huang, K., Dong, X. and Zhuang, B., Li, S., Ge, B., Wu, Q., Cheng, Y., Wang, Y., Lee, H.-J., Kim, C.-H., Fu, J. S., Wang, T., Chin, M., Woo, J.-H., Zhang, Q., Wang, Z. and Carmichael, G. R.: Air Quality and Climate Change, Topic 3 of the Model Inter-Comparison Study for Asia Phase III (MICS-Asia III) Part 1: Overview and Model Evaluation. *Atmospheric Chemistry and Physics*, 18, 4859-4884, 2018.

[20] Song, S., **Gao, M.**, Xu, W., Shao, J., Shi, G., Wang, S., Wang, Y., Sun, Y. and McElroy, M. B.: Fine Particle pH for Beijing Winter Haze as Inferred from Different Thermodynamic Equilibrium Models. *Atmospheric Chemistry Physics*, 18, 7423-7438, 2018 (**co-first author**). ***(ESI Highly Cited Paper)***

[19] Lennartson, E., Wang, J., Garcia, L., Ge, C., Carmichael, G. R., **Gao, M.**, Kim, J. and Scott, J.: Diurnal Variation of Aerosol Optical Depth and $PM_{2.5}$ in South Korea: A Synthesis from AERONET, Satellite (GOCI), KORUS-AQ Observation, and WRF-Chem model. *Atmospheric Chemistry and Physics*, 18, 15125-15144, 2018.

[18] Lu, X., Hong, J., Zhang, L., Cooper, O. R., Schultz, M. G., Xu, X., Wang, T., **Gao, M.**, Zhao, Y. and Zhang, Y.: Severe Surface Ozone Pollution in China: A Global Perspective. *Environmental Science & Technology Letters*, 5(8), 487-494, 2018. ***(ESI Top 0.1% Hot Paper and Highly Cited Paper)***

[17] Pan, Y., Tian, S., Liu, D., Fang, Y., Zhu, X., **Gao, M.**, Gao, J., Wentworth, G., Michalski, G. and Wang, Y.: Source Apportionment of Aerosol Ammonium in an Ammonia Rich Atmosphere: An Isotopic Study of Summer Clean and Hazy Days in Urban Beijing. *Journal of Geophysical Research: Atmospheres*, 123(10), 5681-5689, 2018.

[16] Pan, Y., Tian, S., Liu, D., Fang, Y., Zhu, X., **Gao, M.**, Gao, J., Michalski, G. and Wang, Y.: Isotopic Evidence for Enhanced Fossil Fuel Sources of Aerosol Ammonium in the Urban Atmosphere. *Environmental Pollution*, 238, 942-947, 2018.

[15] Lu, X., Zhang, L., Liu, X., **Gao, M.**, Zhao, Y. and Shao, J.: Lower Tropospheric Ozone over India and Its Linkage to the South Asian Monsoon. *Atmospheric Chemistry and Physics*, 18, 3101-3118, 2018.

[14] Ji, D., Yan, Y., Wang, Z., He, J., Liu, B., Sun, Y., **Gao, M.**, Li, Y., Cao, W., Cui, Y. and Hu, B.: Two-year Continuous Measurements of Carbonaceous Aerosols in Urban Beijing, China: Temporal Variations, Characteristics and Source Analyses. *Chemosphere*, 200, 191-200, 2018.

2017

[13] **Gao, M.**, Liu, Z., Wang, Y., Lu, X., Ji, D., Wang, L., Li, M., Wang, Z., Zhang, Q. and Carmichael, G. R.: Distinguishing the Roles of Meteorology, Emission Control Measures, Regional Transport, and Co-benefits of Reduced Aerosol Feedbacks in APEC Blue. *Atmospheric Environment*, 167, 476-486, 2017.

[12] **Gao, M.**, Saide, P. E., Xin, J., Wang, Y., Liu, Z., Wang, Y., Wang, Z., Pagowski, M., Guttikunda, S. K. and Carmichael, G. R.: Estimates of Health Impacts and Radiative Forcing in Winter Haze in Eastern China through Constraints of Surface $PM_{2.5}$ Predictions. *Environmental Science and Technology*, 51(4), 2178-2185, 2017.

[11] Liang, F., **Gao, M.**, Xiao, Q., Carmichael, G. R., Pan, X. and Liu, Y.: Evaluation of a Data Fusion Approach to Estimate Daily $PM_{2.5}$ Levels in North China. *Environmental Research*, 158, 54-60, 2017.

[10] Liao, Z., **Gao, M.**, Sun, J. and Fan, S.: The Impact of Synoptic Circulation on Air Quality and Pollution-related Human Health in the Yangtze River Delta Region. *Science of the Total Environment*, 607, 838-846, 2017.

[9] Wang, L., Ji, D., Li, Y., **Gao, M.**, Tian, S., Wen, T., Liu, Z., Wang, L., Xu, P., Jiang, C. and Wang,

Y.: The Impact of Relative Humidity on the Size Distribution and Chemical Processes of Major Water-soluble Inorganic Ions in the Megacity of Chongqing, China. *Atmospheric Research*, 192, 19-29, 2017.

2016

[8] **Gao, M.**, Carmichael, G. R., Wang, Y., Ji, D., Liu, Z. and Wang, Z.: Improving Simulations of Sulfate Aerosols during Winter Haze over Northern China: The Impacts of Heterogeneous Oxidation by NO_2. *Frontiers of Environmental Science and Engineering*, 10(5), 16, 2016.

[7] **Gao, M.**, Carmichael, G. R., Saide, P. E., Lu, Z., Yu, M., Streets, D. G. and Wang, Z., Response of Winter Fine Particulate Matter Concentrations to Emission and Meteorology Changes in North China. *Atmospheric Chemistry and Physics*, 16(18), 11837, 2016.

[6] **Gao, M.**, Carmichael, G. R., Wang, Y., Saide, P. E., Yu, M., Xin, J., Liu, Z. and Wang, Z.: Modeling Study of the 2010 Regional Haze Event in the North China Plain. *Atmospheric Chemistry and Physics*, 16(3), 1673-1691, 2016.

[5] Cheng, Y., Zheng, G., Wei, C., Mu, Q., Zheng, B., Wang, Z., **Gao, M.**, Zhang, Q., He, K., Carmichael, G. R. and Pschl, U.: Reactive Nitrogen Chemistry in Aerosol Water as a Source of Sulfate during Haze Events in China. ***Science Advances***, 2(12), e1601530, 2016. ***(ESI Top 0.1% Hot Paper and Highly Cited Paper)***

Before 2016

[4] **Gao, M.**, Guttikunda, S. K., Carmichael, G. R., Wang, Y., Liu, Z., Stanier, C. O., Saide, P. E. and Yu, M.: Health Impacts and Economic Losses Assessment of the 2013 Severe Haze Event in Beijing Area. *Science of the Total Environment*, 511, 553-561, 2015. ***(ESI Highly Cited Paper)***

[3] **Gao, M.**, An, J., Hang, Y. and Li, Y.: Comparison of PMF and PCA/APCS for VOCs Source Apportionment in North Suburb of Nanjing. *Journal of Meteorology and Environment*, 30, 43-50, 2014.

[2] Lei, D., Jin, L., Zhu, Z. and **Gao, M.**: Advances in the Study of Water Vapor Vertical Transport into Stratosphere by Deep Convections. *Journal of NUIST: Natural Science Edition*, 4, 241-247, 2012.

[1] Yang, S., Ma, J., Hu, Z., Jin, L., Xu, X. and **Gao, M.**: Influence of Aerosols Mixing State on the Microphysics of Warm Clouds in North China. *Transactions of Atmospheric Sciences*, 34, 606-613, 2011.

Book Chapter

[1] **Gao, M.**, Carmichael, G., Wang, Y., Saide, P., Liu, Z., Xin, J., Shan, Y. and Wang., Z.: "Chemical and Meteorological Feedbacks in the Formation of Intense Haze Events" in *Air Pollution in Eastern Asia: An Integrated Perspective* (ISSI Scientific Report Series), Springer, 1st ed. 2017 edition.

Thesis

[1] **Gao, M.**: Improving Understanding of Haze Pollution in the North China Plain via Atmospheric Modeling and Data Assimilation, University of Iowa, 2015

Invited Presentations

1. 10/11/2019, China's Clean Air Action Has Suppressed Unfavorable Influences of Climate on Wintertime $PM_{2.5}$ in Beijing, Workshop of State Key Joint Laboratory of Environment Simulation and Pollution Control, Beijing
2. 17/10/2019, Air Pollution in Asia and its Interactions with Health and Climate, Geography and Resource Management Seminar Series, Chinese University of Hong Kong
3. 19/5/2020, 氣溶膠生成與天氣、氣候相互作用數值模擬, 環境科學與工程學院, 南京

信息工程大學

4. 05/31/2019, Invited speaker, 4/5/2020, Aerosol Pollution in Asia and its Interactions with Climate, EGU General Assembly 2020
5. **Hong Kong Polytechnic University Department of Civil and Environmental Engineering Seminar**: Feeling the Pulse of the Ocean, Hong Kong, China.
6. 04/11/2019, **Max Planck Institute for Chemistry Seminar**: Seasonal Prediction of Indian Wintertime Aerosol Pollution using the Ocean Memory Effect, Mainz, Germany.
7. 03/05/2019, **USTC**: Seasonal Prediction of Indian Wintertime Aerosol Pollution using the Ocean Memory Effect, Mainz, Germany.
8. 10/12/2018, **Nanjing University of Information Science and Technology Atmospheric Physics Seminar**: Secular Decrease of Wind Power Potential in India Associated with Warming Indian Ocean, Nanjing, China.
9. 09/26/2018, **City University of Hong Kong School of Energy and Environment Seminar**: Secular Decrease of Wind Power Potential in India Associated with Warming Indian Ocean, Kowloon, Hong Kong.
10. 09/27/2018, **Hong Kong Baptist University Department of Geography Public Seminar Series**: Secular Decrease of Wind Power Potential in India Associated with Warming Indian Ocean, Hong Kong.
11. 03/18/2018, **Peking University Excellent Youth Talent International Forum**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Beijing, China.
12. 07/30/2017, **Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XIV)**: Improving the Understanding of Haze Pollution in the North China Plain via Atmospheric Modeling and Data Assimilation, New York, USA.
13. 04/17/2017, **Tsinghua University Earth System Sciences Seminar**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Beijing, China.
14. 04/12/2017, **State Key Laboratory of Atmospheric Boundary Layer Physics and Atmospheric Chemistry (LAPC), Chinese Academy of Sciences Seminar**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Beijing, China.
15. 03/28/2017, **Nanjing University School of Environment Seminar**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Nanjing, China.
16. 03/23/2017, **Sun Yat-Sen University Atmospheric Sciences Seminar**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Guangzhou, China.
17. 03/22/2017, **State Key Laboratory of Organic Geochemistry (SKLOG), Chinese Academy of Sciences Seminar**: Haze Pollution in Northern China from a Modeling Perspective, Guangzhou, China.
18. 01/20/2017, **Yale University**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, New Haven, CT, USA.
19. 05/02/2016, **Harvard University China Project Seminar**: Haze Pollution in Northern China from a Modeling Perspective, Cambridge, MA, USA.
20. 03/22/2016, **Yonsei University Atmospheric Sciences Seminar**: Improving the Understanding of Haze Pollution in the North China Plain via Atmospheric Modeling and Data Assimilation, Seoul, South Korea.
21. 03/18/2016, **Nanjing University of Information Science and Technology Atmospheric Physics Seminar**: Haze Pollution in the North China Plain, Nanjing, China.
22. 03/01/2016, **International Space Science Institute Workshop on Persistent Regional Air Pollution in Asia**: Chemical and Meteorological Feedbacks in the Formation of Intense Haze Events, Beijing, China.
23. 07/06/2015, **World Meteorological Organization (WMO) Near Real Time (NRT) Data Application Pilot Project Workshop**: Modeling Study of the 2010 Regional Haze Event in the North China Plain, Beijing, China.

Grants as Principal Investigator

[1] 排放與氣候因素對近十多年北京冬季 PM2.5 濃度變化的影響, 環境模擬與污染控制國家重點

聯合實驗室（清華大學）開放課題(19K03ESPCT)，2019-2021, 80,000 CNY

[2] 中國與印度重污染區域臭氧的季節性與污染源特徵, 中國氣象局氣溶膠與雲降水重點開放實驗室開放課題 (KDW1901)，2019-2021, 30,000 CNY

[3] APEC 期間多污染物減排對黑碳光吸收影響的模擬研究，江蘇省大氣環境監測與污染控制高技術研究重點實驗室開放基金(KHK1902)，2019-2021, 50,000 CNY

[4] 排放與氣候因素對近十多年廣東冬季 PM2.5 濃度變化的影響， 廣東省自然科學基金-面上項目(2019A1515011633)，2019-2022, 100,000 CNY

[5] Assimilating surface PM2.5 and ozone measurements to improve health exposure assessment and air quality forecasting in South China, Early Career Scheme (ECS) (22201820), 2021-2023, 579,972 HKD

[6] Co-occurrence and Exposure of Heat and Haze Hazards over China under a Warming Climate, Tier 2 Start-up Grant (SGT2) 2019/20, 499,000 HKD

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Service

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- **Proposal Reviewer:** *Harvard CHAN-NIEHS Center for Environmental Health Pilot Project*
- **Conferences Session Convener and Chair:** *AGU, AOGS, etc.*
- **International Project Working Group Leader:** *The Model Inter-Comparison Study for Asia (MICS-Asia) Phase III Topic 3: Air Quality and Climate Change*